

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

--

CLASS

--

INDEX
NUMBER

--	--	--	--

BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

11 September 2017

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

Circle the question number of the question attempted.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
4 / 5	
Total	

This document consists of **22** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

- 1 The *lac* operon is an operon required for the uptake and metabolism of lactose in *Escherichia coli* and many other bacteria. Glucose is the preferred carbon source for most bacteria, as glucose requires fewer steps and less energy to break down than lactose. However, if lactose is the only sugar available, the *E. coli* uses it as an energy source and the *lac* operon allows for the effective digestion of lactose when glucose is not available.

To use lactose, the bacteria must express the *lac* operon genes, which encode key enzymes for lactose uptake and metabolism. Fig. 1.1 shows the results of an experiment carried out to determine the effects of adding lactose on the expression of some of the genes involved in the breakdown of lactose.

The initial gene expression was measured by determining the mRNA produced at time 0. This was taken as 10% (black bars). All the other values are relative to this initial value taken every 30 seconds over the next 2 minutes.

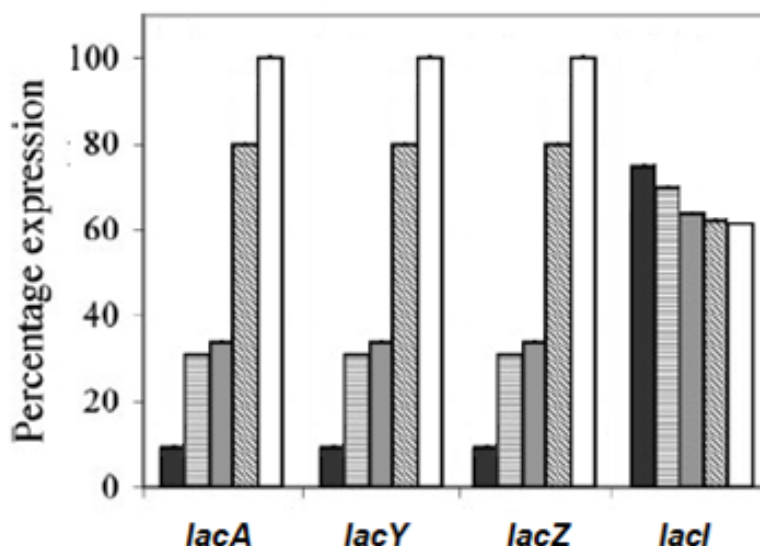


Fig. 1.1

(a) Suggest why operons are necessary in bacteria. [2]

1. Operons allow for the simultaneous regulation of related genes of related functions which are involved in the same metabolic activity;;
2. Genes coding for the proteins/enzymes of a single biochemical pathway / of related functions / same metabolic activity, (such as the catabolism of carbohydrates) are grouped together into an operon for easier control;;
OR
3. Operons allow/enable the simultaneous regulation of related genes to adapt/in response to environmental changes;;
4. The genes on the operons are only expressed when required / the bacteria only produce the proteins/enzymes that are required to increase efficiency / conserve/prevent the waste of energy and resources;;
5. Thus the bacteria are at a selective advantage;;

(b) Using data from *lacA*, *lacY* and *lacZ* in Fig. 1.1 and your knowledge of how different types of operons are regulated, explain how lactose is able to control the expression of these genes. [4]

1. Gene expression increased from 10% at 0 min to 100% at 2 min;;
2. Operon is turned on / need to be turned on / a inducible system where expression of *lacA*, *lacY* and *lacZ* are inactive / not constitutively active (10% at time 0);;
3. lactose binds to the repressor (and act as a inducer) / allolactose binds to the allosteric site of the lac repressor, the lac repressor protein changes to inactive conformation, thus the repressor protein is unable to bind to the operator;;
4. RNA polymerase is able to recognise and bind to the promoter, and transcription of the (structural) genes of the operon led to formation of mRNA;;
(resulting in an increase in gene expression)

It has been suggested that not all genes involved in lactose hydrolysis are organised into one single operon.

(c) Use evidence from Fig. 1.1 to support the statement above. [3]

1. *lacA*, *lacY* and *lacZ* respond to lactose (inducer) to the same extent;;
2. percentage expression for *lacA*, *lacY* and *lacZ* increases from 10% at 0s to 100% at 2 min;;
3. suggesting that *lacA*, *lacY* and *lacZ* are under the control of the same promoter and operator;;
OR
4. *lacI* does not respond significantly to lactose in the same period of time;;
5. percentage expression for *lacA*, *lacY* and *lacZ* increases from 10% at 0s to 100% at 2 min compared to *lacI* which decreases from 80% at 0s to 60% at 2 min (data tbu);;
6. suggesting that *lacI* must be found in a different region of the bacterial chromosome, under the control of a different promoter;;
OR
7. Gene expression of *lacA*, *lacY* and *lacZ* increase whereas *lacI* decrease;;
8. percentage expression for *lacA*, *lacY* and *lacZ* increases from 10% at 0s to 100% at 2 min compared to *lacI* which decreases from 80% at 0s to 60% at 2 min (data tbu);;
9. suggests that *lacI* is not transcribed simultaneously, *lacI* may be controlled by a different promoter;;
Mark once for promoter

(d) A series of mutations was introduced into the *lac* operon, resulting in the inversion of the operator and the promoter regions.

Suggest the effect on the transcription of the *lac* genes when lactose is absent. [2]

1. In the absence of lactose, lac repressor protein binds to the operator region, which is now upstream of the promoter / lac repressor protein does not physically block RNA polymerase;;
Accept ref to the effect of the inversion;;
2. Hence, RNA polymerase can bind to the promoter, transcription of *lac* genes is now always on / cannot be turned off;;

Owing to *E. coli*'s rapid growth rate, *E. coli* has been an expression host of choice in the biotechnology industry for large-scale production of anti-freeze proteins (AFPs). AFPs is a class of polypeptides that help to stop ice forming inside the Arctic and Antarctic fishes thus permitting their survival in sub-zero environments.

In the Arctic and Antarctic, environmental temperatures can reach low to freezing levels. These fishes indigenous to these habitats are presented with potential desiccation, which can lead to potentially detrimental challenges such as decreased enzymatic rates and freezing. Besides hindering cellular processes, sub-zero temperatures induce ice crystals formation, which can lead to cell death by rupturing cells either physically or through osmotic pressure changes.

Commercially, there appears to be countless applications for AFPs:

- as additives to frozen foods to lengthen the shelf life
- incorporation with the genome of the raw foods to retard ice crystal growth
- to prevent damage to agricultural crops by increasing freeze tolerance of crop plants and extending the harvest season in cooler climates
- introduction into ice cream and yogurt products to allow the production of very creamy, dense, reduced fat ice cream with fewer additives.

(e) Outline how the genome of *E. coli* and the genome of the fish are similar and how they are different. [4]

Similar:

1. both have double-stranded DNA;

Differences (max 3)

Feature		Prokaryotic	Eukaryotic	
1.	Linearity/circularity	Circular / Looped chromosome	Linear chromosomes	;;
2.	Association with proteins	Naked / Associated with H-NS proteins (nucleoid-associated proteins)	Associate with histones and scaffolding proteins	;;
3.	Introns	Absence of introns	Presence of introns	;;
4.	Genome Size / Number of genes	Small / Fewer genes present	Large / Many more genes present	;;
5.	Number of chromosomes	Single chromosome	Multiple chromosomes	;;
6.	Origin of replication	One per chromosome	Many per chromosome	
7.	Operons	Presence of operons	Absence of operons	

AVP: (pt 5-7) [max 1]

Extra: prokaryotic, in cytoplasm / not membrane bound vs eukaryotic, membrane bound/in nucleus;

(f) Anti-freeze glycoprotein (AFGP) is one type of anti-freeze protein. Messenger RNA coding for AFGP is translated at a ribosome to produce a polypeptide. Describe how this polypeptide is then processed to make AFGP. [4]

1. In the rER, the AFGP polypeptide will coil and fold into (the geometrically regular) secondary structures/ α -helix and β -pleated sheet (held together by hydrogen bonding between C=O and –NH groups of the amino acids);;

OR

The AFGP polypeptide will undergo further bending, coiling, folding, to form the (specific) tertiary structure/3D conformation;;

2. biochemical modification/ glycosylation takes place / AFGP may be modified by enzymes in the ER lumen that add carbohydrate chains to them;;

OR

3. the protein is released into the lumen of the Golgi body for further modification, sorting and packaging into vesicles;;

4. The AFGP is packaged into transport vesicles / Golgi vesicles and transported towards the Golgi body / other parts of the cell;;

5. movement from rER to GA;;

Some fish produce another anti-freeze protein, called AFP II. The tissues of these fish were tested for the presence of AFP II and the mRNA coding for AFP II. The results are shown in Table 1.1.

Table 1.1

molecule	present in
AFP II protein	all tissues
AFP II mRNA	liver tissue only

(g) Explain the distribution of the AFP II protein and AFP II mRNA. [4]

- AFP II gene/allele is activated only in liver cells / deactivated in cells other than liver cells;;**
ALLOW "switched on/off"
- activation could be due to DNA demethylation/histone acetylation/activator protein binding to enhancer;;**
OR
- deactivation could be due to DNA methylation/histone deacetylation/repressor protein binding to silencer;;**
- transcription and translation/protein synthesis of AFP II occurs/takes place only in liver cells;;**
Ref to liver cells required only once if context / chain of argument is clear.
- the protein/AFP II is secreted from liver cells / transported around the body, presence of protein;;**
- AFP II in all tissues prevents freezing/ice forming in all parts of the body;;**

(h) With reference to named examples, describe the roles performed by proteins involved in transport in fishes. [2]

- haemoglobin/ myoglobin, for oxygen transport/ storage;;**
- membrane carriers/ channels, with role in passive transport;;**
- example of carrier/ channel protein, with specific role;;**
 - GLUT/glucose transporter for glucose transport / aquaporin for water molecules transport / H⁺ channel in stalked particle**
- membrane pumps/ AW with role in active transport;;**
- example of pump/ AW with role;;**
 - Na⁺/K⁺ pump for transporting Na⁺ (out of the cell) and K⁺ (into the cell), H⁺ pump along ETC**
- further example of membrane transport protein with contrasting role;;**
- electron carriers/electron transport chain, components for chemiosmosis/ ATP synthesis/redox reactions;;**

Table 1.2 shows Earth's ice ages over the last 850 million years.

Table 1.2

Ice age	Time / millions of years ago
Quaternary	0 to 2.6
Karoo	260 to 360
Andean-Saharan	420 to 460
Cryogenian	630 to 850

Fig. 1.2 shows how the number of families of fishes has changed over time.

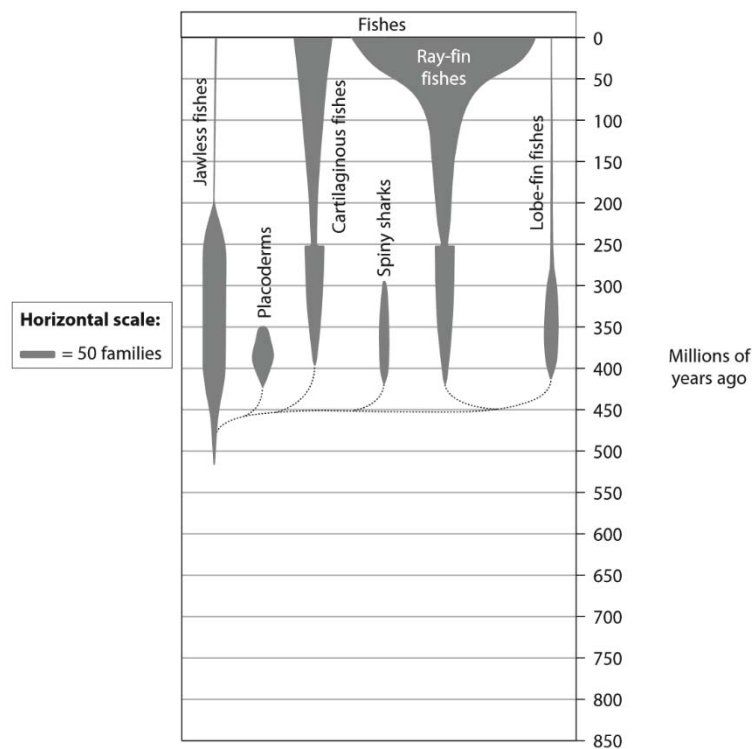


Fig. 1.2

- (i) Many different types of AFPs are produced by ray-fin fishes. Analyse the data to explain when these ray-fin fishes are likely to have evolved the ability to produce AFPs.
[2]

1. (sea) ice is a selection pressure for AFPs / AFPs are advantageous (only) when there is (sea) ice;;
ALLOW AFPs allow fish to survive the ice age
OR
2. so AFPs are likely to have appeared/increased in frequency during an ice age;;
OR
3. the only ice ages since the existence of the ray-fin fish are the Quaternary and Karoo;;
4. therefore ray-fin fish producing AFPs are likely to have evolved in the last 2.6 million years / between 260 and 360 million years ago;;
ALLOW during the Karoo / Quaternary (ice age)

[Total: 27]

- 2** Measles is a highly contagious, serious disease caused by *Morbillivirus*, a single-stranded enveloped RNA virus. Envelope glycoproteins mediate transmission of the virus into host cells in the human respiratory tract. Once inside the host cell, the viral RNA genome is transcribed into mRNA, which undergoes translation to manufacture viral proteins. These viral proteins function to form capsid proteins for new viruses which eventually leave the host cell.

(a) With reference to the information given, outline how viruses challenge the concept of what is considered living. [2]

- 1. Viruses are acellular and do not contain cytoplasm or cellular organelles;;**
- 2. Viruses contain only one type of hereditary material / either DNA or RNA but never both;;**
- 3. Viruses must replicate using the host cell's (metabolic machinery) ribosomes, amino acids, enzymes, nucleotides, energy – any 1 e.g.;;**
OR
- 4. Outside of host cells, viruses do not carry out any metabolism / grow or divide;;**
OR
- 5. The new viral components are synthesised and assembled within the infected host cell;;**
(any 2)

In the 1980s, measles caused an estimated 2.6 million deaths each year, and the disease remains one of the leading causes of death among young children globally.

The number of cases of measles is reported to the World Health Organisation (WHO) by countries throughout the world so that global data is collected.

Fig. 2.1 shows the global data collected between January 2008 and December 2012.

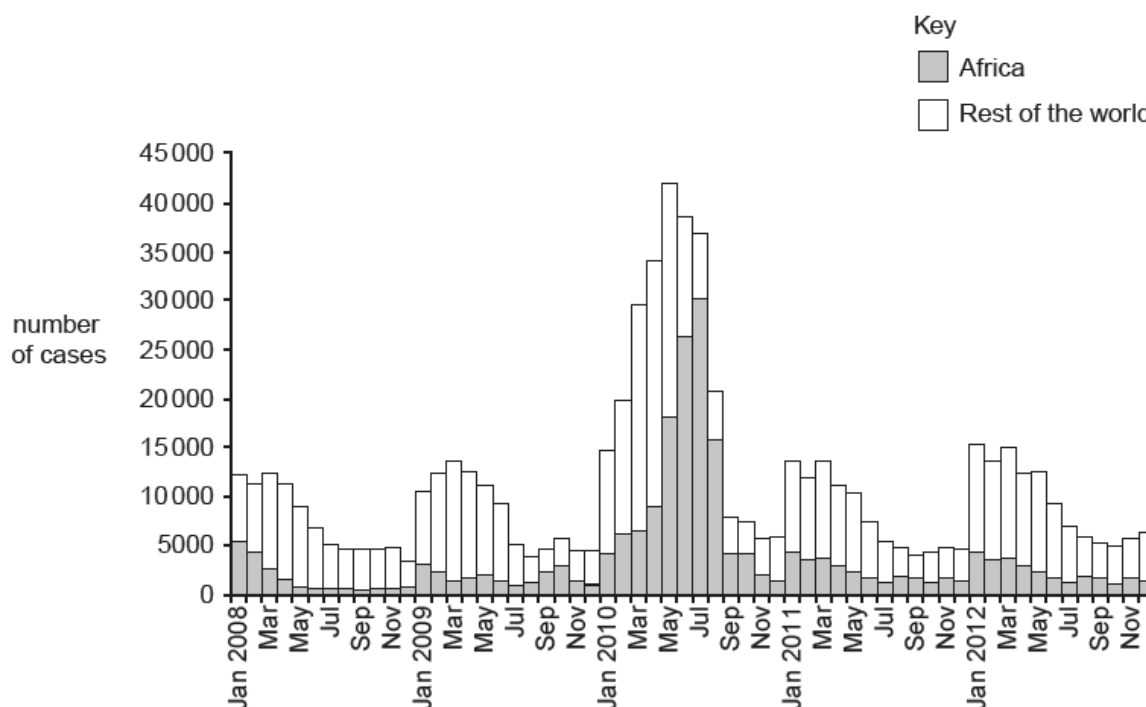


Fig. 2.1

(b) Use the data in Fig. 2.1 to describe the pattern shown in the number of cases of measles reported to the WHO between January 2008 and December 2012. [3]

1. Number of cases fluctuated between 2008 to 2012 / in all years;;
 2. Number of cases tends to be higher at the beginning of each year (than at the end of each year), except in 2010;;
 3. Number of cases in the rest of the world are greater than in Africa each year (or vice versa), except in 2010;;
 4. Number of cases (much) higher in 2010;;
 5. with the highest peak/highest total number of *quote number between 41 875 - 42 143 cases in (May) 2010;;*
- OR
6. with the highest number of 30 000 cases in Africa in (July) 2010;;
 7. Epidemic lasted longer in 2010;;
 8. There has been 5 (accept: 4 since no data before Jan 2008) outbreaks/epidemics between Jan 2008 and Dec 2012;;

- (c) Routine measles vaccination for children, combined with mass immunisation campaigns in countries with high case and death rates, are key public health strategies to reduce global measles deaths. By 2016, about 85% of the world's children received one dose of the measles vaccine by their first birthday, and the global push to improve vaccine coverage resulted in a 79% reduction in deaths.

(i) State precisely the type of immunity gained by receiving a measles vaccine. [1]

1. (Adaptive) Acquired active immunity;;

(ii) Outline one benefit of vaccination. [1]

Benefits of Vaccination

- 1. Protect the individual against disease by conferring immunity to the individual without prior exposure to a specific pathogen;;**
- 2. Confer lifelong immunity to the individual by providing secondary immune response against subsequent encounter with the same pathogen;;**
- 3. Confer herd immunity to unvaccinated individuals in a community (e.g. pregnant women and people with allergies/weakened immune system), reducing possibility of transmission between individuals / unvaccinated individuals have a very low risk of becoming infected;;**
- 4. Contribute to the elimination / eradication of infectious diseases within human population (if pathogen relies only on human as host);;**

(iii) Outline one risk of vaccination. [1]

Risks

- 1. Side effects/adverse reactions after vaccination have been observed in small numbers of individuals (e.g. life-threatening allergic reaction, fainting and rashes etc.);;**
- 2. Some individuals are more susceptible to vaccination risks than others (e.g. individuals with weakened immune systems);;**
- 3. (for live attenuated vaccines) Pathogen used in vaccines may regain its virulence and cause disease;;**

- (d) Unlike measles for which an effective vaccine has been developed, it has been extremely difficult to design an effective vaccine against malaria. Malaria is a disease caused by the parasite *Plasmodium falciparum*. *P. falciparum* multiplies in liver cells of the host before emerging after 9-30 days wrapped in the liver cell surface membrane. They enter red blood cells, multiply and then cause rupture of the host cells, resulting in the release of more parasites every 36-48 hours, in a manner that has some similarity to that of viruses.

Use the information given and your own knowledge to suggest why it has been extremely difficult to design an effective vaccine against malaria. [2]

1. The parasite mostly stays inside host cells, thus evading the host immune system;;
2. The parasite mostly stays inside host cells/uses liver cell membrane as 'covering' and is therefore disguised as 'self/non-foreign';;
3. Difficulty in designing a vaccine which stimulates both cell-mediated and humoral responses to bring about parasite clearance and provide future immunity against the parasite;;
4. Diversity/large degree of variation/change in parasite antigens such that existing antibodies targeting the old antigens bind poorly/cannot recognise and bind to the new antigenic sites, allowing parasites with new antigens to evade immunological memory against the original parasite;;

(Any 2)

- (e) Another infectious disease, Tuberculosis (TB), is one of the top ten causes of death worldwide. Name the bacterium that causes TB and describe how TB is transmitted. [3]

1. *Mycobacterium tuberculosis*;; (penalise for spelling)
2. The bacteria are transmitted from person to person through fine aerosol droplets;; (accept: airborne)
3. formed when an infected person with the active disease sneezes/coughs/breathes (accept: spits/talks);;
OR
4. droplets are inhaled by an uninfected person;;

[Total: 13]

- 3** Dengue fever is a disease spread by a particular species of mosquito, *Aedes aegypti*. The incidence of this disease and the numbers of this species of mosquito have increased dramatically in recent years, spreading beyond the tropics. This has been attributed to global warming.

(a) Explain how global warming has resulted in the spread of dengue beyond the tropics.
[2]

- 1. Global warming results in increase in geographical/distribution range of *A. aegypti* as they have increased survival in its new location (that used to be too cold) / are extending increasingly into temperate zones;;**
- 2. Also results in increase in number of mosquito vectors as higher temperatures lead to increased metabolism / accelerates their development;;**
OR
- 3. Increases activity of female mosquitoes and reduce the incubation time for them to become infectious / transmit DENV;;**
OR
- 4. Rate of viral replication within vector will increase and extrinsic incubation period (before DENV becomes transmissible to another host) will shorten;;**

In an attempt to reduce the numbers of *A. aegypti*, male mosquitoes infected with the *Wolbachia* bacteria have been produced and released into the wild to mate with females. *Wolbachia* naturally occurs in up to 60% of all insect species, but not in *A. aegypti*. *Wolbachia* induces a conditional sterility that occurs within the mosquitoes due to cytoplasmic incompatibility, shown in Fig. 3.1, a concept first introduced in a paper published by Dr. Hannes Laven in 1967.

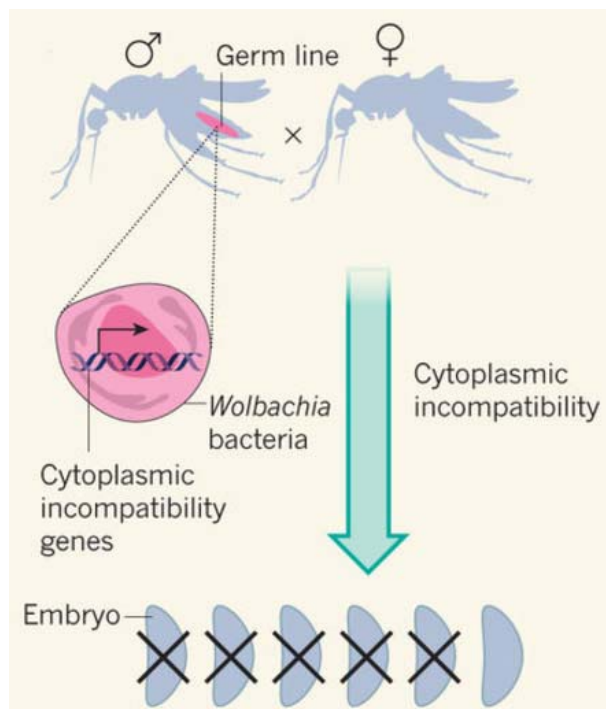


Fig. 3.1

(b) The cytoplasmic incompatibility genes are DNA in nature. DNA is a double helix consisting of two polynucleotide strands held together by phosphodiester bonds between the adjacent nucleotides. Each strand contains a sugar-phosphate backbone and hydrogen bonds are formed between the complementary strands via complementary base pairing.

Describe two other structural features of DNA. [2]

1. **The two strands coil around each other in a right-handed double helix;;**
2. **The strands are antiparallel/run in opposite directions/one strand runs in the 5' to 3' direction while the complementary strand runs in the 3' to 5' direction;;**
3. **Each nucleotide comprising of a deoxyribose, a phosphate group and one of the four nitrogenous bases - Adenine, Thymine, Cytosine or Guanine;;**
4. **The nitrogenous bases are arranged as side groups of the chains (oriented toward the central axis);;**
5. **The width between the 2 sugar-phosphate backbones is constant at 2nm, equals to the width of 1 base pair i.e. 1 purine + 1 pyrimidine;;**
6. **One complete turn of the double helix measures 3.4nm in length and comprises 10 base pairs;;**

There are three possible matings between the male and female mosquitoes in the wild as shown in Table 3.1.

Table 3.1

	male	female	results
cross 1	infected	uninfected	lay eggs that are not viable and do not hatch
cross 2	infected	infected	infected offspring
cross 3	uninfected	infected	infected offspring

Embryonic development aborts when sperm from an infected male fertilises an uninfected egg and the paternal genome does not contribute to the development of the embryos that are not viable.

(c) Based on the information provided and Cross 1, suggest how *Wolbachia* induces sterility. [1]

1. **Cytoplasmic incompatibility genes in *Wolbachia* are transcribed and translated to form proteins / *Wolbachia* secretes a (DNA binding) protein that binds to the sperm/paternal DNA/chromosome;;**

In 1970, Erich Jost repeated Laven's earlier work.

(d) Explain why repeating the work of others is an important part of science research. [2]

1. **To test/check results/findings;;**
2. **(if results support) to build scientific consensus / increase confidence (A: reliability, R: accuracy) in the findings;;**
3. **(if results different) to revise/refine the theory/model/hypothesis;;**

In 2016, hundred thousands of male mosquitos with *Wolbachia* were released at 3 selected sites in Singapore: Braddell Heights, Nee Soon East, and Tampines West.

(e) State why releasing such large numbers of male mosquitoes did not immediately increase the risk of transmission of dengue fever in these estates. [1]

1. **Only females spread dengue;;**
2. **Males do not bite / feed on blood;;**

Another method employed in Australia involves the release of both male and female mosquitoes with *Wolbachia* into the wild. An advantage of this method is that there is no need for further releases of mosquitoes with *Wolbachia*.

(f) With reference to Table 3.1, explain why there is no need for further releases with this method. [2]

1. **Infected female mosquitoes can mate with both uninfected and infected males to produce infected offspring;;**
2. **Passing the bacteria from generation to generation;;**
3. **Over time, the percentage of mosquitoes carrying *Wolbachia* grows until it remains high without the need for further releases;;**

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 (a) Describe the reproductive cycle of the influenza virus and explain how new strains of the virus may arise as a result of mutation. [13]

Adsorption

1. The haemagglutinin glycoproteins (on the viral envelope) recognise and bind to (sialic acid containing) receptors on the host cell plasma membrane;;

Penetration and Uncoating

2. The influenza virus enters the host cell by receptor-mediated endocytosis;;
3. whereby the host cell membrane invaginates (and pinches off), engulfing / placing the virus in an endocytic vesicle;;
4. Acidification causes the viral envelope to fuse with the endocytic vesicle membrane;;
5. releasing the nucleocapsid into the cytoplasm of the cell;;
6. The capsid is then degraded by cellular enzymes, releasing the viral nucleic acid;;

Replication

7. The viral genome functions as the template for synthesis of complementary RNA strand (cRNA);;
8. using the viral RNA-dependent RNA polymerase (and host RNA nucleotides in the nucleus);;
9. This cRNA acts as a template for the synthesis of new copies of viral RNA (genome);;
10. and functions as mRNA which undergoes translation to produce viral proteins (e.g. capsid proteins and glycoproteins for the viral envelope);;

Maturation

11. The glycoproteins synthesised (by ribosomes on the ER) are transported to the (host cell's) plasma membrane via vesicles and incorporated into the plasma membrane;;
12. The capsid is assembled around the viral RNA genome and the RNA-dependent RNA polymerase;;

Release

13. The new influenza viruses bud off from the host cell's plasma membrane / ref. to budding, resulting in the new viruses acquiring their lipid bilayer from the plasma membrane of the host cell, with viral proteins and glycoproteins embedded;;
14. Neuraminidase catalyses the cleavage of sialic acid residues from haemagglutinin, facilitating the release of the newly replicated viruses (for the next round of infection);;

[max 11m]

Antigenic drift

15. **Point mutations in haemagglutinin gene** in influenza virus occur during the replication of viral RNA;;
16. **Accumulation of mutations** results in **slight changes** to the **shape of haemagglutinin**;;
17. **leading to antigenic drift**;;
18. **Shape of the new haemagglutinin is now complementary to other membrane receptors found on new types of host cells;;**

QWC

Good spread of knowledge communicated without ambiguity to include:

at least 1 MP from each of the 2 sections – reproductive cycle of virus (pt 1-14) and antigenic drift (pt 15-18);;

(b) Describe the process of transduction and its advantages to prokaryotes. [12]

1. Transduction is the process by which bacterial DNA / genes is transferred from one bacterium (host cell) to another (recipient cell) via a bacteriophage;;
2. Generalised transduction requires infection of a bacterium by a virulent bacteriophage (and any random portion of the bacterial DNA transferred);;
3. During the assembly of the newly replicated phage genome within the phage capsid, a small piece of the host cell's degraded DNA gets mistakenly packaged within the capsid (defective phage);;
4. Specialised transduction requires infection of a bacterium by a temperate bacteriophage (and only bacterial genes adjacent to the integrated prophage transferred);;
5. When a temperate phage enters into lytic cycle from lysogenic cycle / spontaneous induction occurs;;
6. Small region of the bacterial DNA that was adjacent to the prophage is excised;;
7. The phage-host hybrid DNA is packaged within a capsid (defective phage);;
8. The defective phages infect / attach to other bacterial cells and inject the piece of host bacterial DNA into the newly infected bacterial cell cytoplasm;;
9. (Generalised & Specialised transduction) Foreign bacterial DNA is incorporated into the recipient / bacterial cell's chromosome through homologous recombination;;
OR
10. (Generalised & Specialised transduction) If there is sufficient homology between the DNA fragments and bacterial chromosome, crossing over occurs, and segments of the original chromosomal DNA will be replaced;;
11. (Specialised Transduction) Phage-host hybrid DNA integrates into the recipient cell's chromosome, as phage enters the lysogenic cycle;;

Advantages

- 12. New alleles can then be incorporated into the bacterial genome of recipient cell;;**
- 13. Recipient cell will subsequently express new characteristics / show change in phenotype;;**
- 14. Generate genetic diversity / variation;;**
- 15. (Example of new allele) Gain antibiotic resistant alleles from other bacteria;;**
- 16. (How it confers selective advantage) Antibiotics less effective against the bacteria;;**
- 17. Thus the bacteria is at a selective advantage;;**

QWC:

scientific argumentation exemplified by:

two or more advantages to prokaryotes (pt 12-17) linked coherently to the correct stage of the process;;

5 (a) Describe the roles of the proteins involved in the process of DNA replication and compare the advantages of PCR with the advantages of DNA replication. [13]

1. Helicase causes the DNA molecule to unwind and unzip;;
2. hydrogen bonds between complementary bases break, causing the 2 parental DNA strands to separate;;
3. single-strand DNA binding proteins bind to the 2 separated parental DNA strands;;
4. to stabilise the single-stranded DNA formed;;
5. Primase catalyses the formation of a short RNA primer (– the start of a new strand in the 5' to 3' direction);;
6. DNA polymerase (then binds to the RNA primer and) adds nucleotides to the free 3' end of the RNA primer/existing strand;;
7. DNA polymerase catalyses the formation of phosphodiester bonds between the nucleotides;;
8. RNA nucleotides of all the RNA primers are replaced with DNA nucleotides by another DNA polymerase;;
9. DNA ligase seals the gaps between the DNA fragments;;
10. by catalysing the formation of phosphodiester bonds between adjacent nucleotides (to form a continuous strand);;

Similarities

11. Both processes allow for the production of large amounts of DNA;;
12. PCR can be fully automated as Taq polymerase can withstand high temperatures without being denatured, DNA replication also do not require replacement of enzymes / OWTTE;;
13. It is easy to set up and use a thermal cycler for PCR, DNA replication also takes place in the nucleus easily;;

Differences

14. PCR has high sensitivity and can amplify sequences from minute amounts of target DNA while replication require the entire template/cannot amplify sequences from minute amounts;;
15. Millions of copies of target DNA can be obtained in a relatively short period of time/in a few hours in PCR while DNA replication requires a longer period/10 to 12 hours;;
16. PCR is robust and can amplify specific sequences from material in which the DNA is badly degraded/embedded in a medium while DNA replication cannot replicate badly degraded DNA;;
17. PCR is a specific process which amplifies only target sequences while DNA replication results in the entire DNA sequence being replicated;;

QWC:

Scientific argumentation exemplified by:

Two or more direct comparisons of the advantage of PCR and/or DNA replication, each clearly set out to show its similarity with, or difference between, **both** PCR and DNA replication;;

(b) Outline the structure of G-protein linked receptor and describe the action of glucagon on liver cells in the regulation of blood glucose concentration. [12]

Structure of G-protein linked receptor

1. G-protein linked receptor consist of a single polypeptide chain coiled and folded into a tertiary structure;;
 2. held together by hydrogen bonds, ionic bonds, disulfide bonds and hydrophobic interactions;;
 3. comprised of seven transmembrane α -helices;;
 4. has different binding sites for signal molecule/ligand and G protein;;
- OR
5. extracellular part/domain of G-protein linked receptor serve as the binding site for (specific) signal molecule/ligand;;
 6. intracellular parts/domain of G-protein linked receptor serves as binding site for G-protein;;
- pt 4 or pt5&6: award once*
7. embedded in and span the plasma membrane, held by weak hydrophobic interactions;;
 8. Non-polar R groups of amino acid residues on the receptor form hydrophobic interactions with non-polar hydrocarbon tails of the membrane phospholipid molecules;;
 9. The extracellular parts of G-protein linked receptors may be glycosylated (as they serve as the binding site for ligands);;

Action of glucagon on liver cells

10. Glucagon recognises and binds to the specific binding site of G-protein linked receptor on the liver cell membrane;;
11. and induces a conformational change in the receptor;;
12. The receptor now binds to G protein and activates it;; (A molecule of GTP replaces the GDP on the G protein)
13. The activated G-protein dissociates from the receptor and activates adenyl cyclase;;
14. Adenyl cyclase catalyses the conversion of ATP to cyclic AMP (cAMP);;
15. The cAMP then acts as a second messenger;;
16. and triggers downstream signalling events/phosphorylation cascade such that glycogen phosphorylase is activated;;
17. Glycogen phosphorylase will catalyse the breakdown of glycogen to glucose/glycogenolysis;;
18. Glucagon also stimulates an increase in the rate of conversion of amino acids and glycerol to glucose/ gluconeogenesis;;
19. so that the blood glucose concentration increases and return back to normal levels.

QWC

Good spread of knowledge communicated without ambiguity to include:

at least 2 MP from each of the 2 sections – structure of G-protein linked receptor (pt 1-9) and action of glucagon on liver cell (pt 10-18);;